

MOOSE 1 NNE, WY, USA

WMO: 725740

Lat: 43.6615N

Lon: 110.7120W

Elev: 6466

StdP: 11.57

Time Zone: -7.00 (NAM)

Period: 04-19

WBAN: 04131

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-16.4	-11.3	-21.0	2.2	-15.9	-16.0	2.9	-11.4	10.9	27.8	10.3	27.5	2.2	N/A	0.619

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	35.0	84.3	54.7	81.7	54.3	78.8	53.7	58.4	74.8	57.0	73.8	55.6	72.9	4.4	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
53.3	77.1	61.8	51.0	70.8	60.8	49.0	65.6	60.1	28.3	74.7	27.2	73.5	26.3	72.7	63.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
9.7	8.4	7.2	-26.2	88.1	5.0	1.6	-29.8	89.2	-32.7	90.1	-35.5	91.0	-39.1	92.2
			-25.8	62.0	5.8	0.9	-29.9	62.6	-33.4	63.1	-36.6	63.6	-40.9	64.3
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	37.2	13.0	17.9	26.6	35.5	45.2	53.6	61.6	59.2	51.5	39.1	26.9	14.9
	DBStd	18.53	11.50	10.54	8.82	7.07	6.27	6.03	4.28	4.42	6.18	7.42	9.88	11.97
	HDD50	5593	1146	898	726	434	171	34	0	4	56	343	692	1089
	HDD65	10165	1611	1318	1191	884	613	342	119	182	406	803	1142	1554
	CDD50	924	0	0	0	1	24	143	361	289	101	5	0	0
	CDD65	18	0	0	0	0	0	1	14	3	0	0	0	0
	CDH74	1920	0	0	0	0	7	226	971	605	109	2	0	0
	CDH80	397	0	0	0	0	0	37	244	106	10	0	0	0
(14) Wind	WSAvg	3.3	3.1	3.9	3.9	3.9	3.7	3.1	3.3	2.8	2.8	3.1	3.1	3.1
(15) Precipitation	PrecAvg	22.0	2.6	1.9	1.9	1.7	2.0	1.9	1.1	1.3	1.4	1.7	2.1	2.8
	PrecMax	30.2	4.6	4.9	4.3	2.5	3.9	4.3	3.2	2.3	3.0	3.1	4.7	7.6
	PrecMin	15.4	0.6	0.7	0.7	0.5	0.4	0.6	0.2	0.3	0.3	0.2	0.4	1.1
	PrecStd	4.4	1.2	0.8	0.9	0.6	1.0	1.1	0.8	0.6	0.6	0.8	1.0	1.4
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	38.7	42.2	52.3	66.8	74.9	84.4	88.2	86.0	81.1	69.7	55.2	41.4
		MCWB	32.2	34.2	39.2	45.2	49.9	55.1	56.0	54.4	54.3	48.0	41.4	32.7
	2%	DB	35.1	38.4	46.2	59.6	70.9	79.9	85.5	82.9	77.1	63.6	50.1	38.0
		MCWB	31.0	32.7	36.7	42.4	48.7	53.9	55.5	53.4	51.9	45.7	39.1	33.3
	5%	DB	32.7	36.2	42.9	54.5	66.5	76.2	83.0	80.4	74.2	59.0	45.4	34.8
		MCWB	29.7	31.7	34.9	40.4	46.9	53.0	55.1	53.0	50.1	43.9	36.4	31.4
	10%	DB	30.2	33.6	39.8	49.4	61.7	72.5	80.0	77.6	70.0	54.0	41.4	32.4
		MCWB	27.8	30.3	33.5	38.1	45.7	51.6	54.7	52.7	49.1	42.0	34.2	29.7
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	33.8	35.4	39.8	47.4	52.0	58.6	61.0	59.4	58.3	50.7	42.4	35.6
		MCDB	36.6	39.8	49.8	64.5	68.5	78.5	76.9	73.9	70.8	65.1	52.3	38.9
	2%	WB	32.4	34.1	37.6	43.7	49.7	55.9	59.1	57.4	54.9	48.1	39.6	33.8
		MCDB	34.8	38.2	45.0	59.1	65.6	75.0	75.5	72.7	68.5	60.3	49.1	36.5
	5%	WB	31.0	32.8	36.0	41.0	47.9	54.2	57.5	56.0	52.9	45.6	37.1	32.0
		MCDB	33.3	36.3	41.7	52.1	63.0	72.7	74.7	72.2	68.3	57.5	43.7	34.5
	10%	WB	28.2	31.5	34.5	38.7	46.0	52.5	56.5	54.6	51.0	43.2	34.6	29.7
		MCDB	30.9	34.6	39.5	48.1	59.5	70.2	74.0	71.7	66.2	52.8	39.7	33.0
(35) Mean Daily Temperature Range	MDBR		19.5	19.3	20.5	21.4	25.8	30.4	35.0	35.1	32.0	25.0	20.2	18.5
	5% DB	MCDBR	17.1	16.0	23.4	32.1	36.6	39.3	40.0	40.6	40.6	35.1	26.1	14.4
		MCWBR	14.5	10.7	13.7	16.7	16.4	16.7	15.9	16.1	17.4	18.5	16.3	11.1
	5% WB	MCDBR	15.2	12.4	19.3	29.2	31.4	35.8	34.7	33.1	32.2	30.1	24.4	13.1
(40) Clear-Sky Solar Irradiance	taub		0.234	0.245	0.261	0.298	0.304	0.311	0.325	0.333	0.301	0.280	0.250	0.232
	taud		2.354	2.313	2.330	2.358	2.489	2.518	2.502	2.461	2.553	2.572	2.516	2.403
	Ebn at Noon		292	306	313	307	306	303	298	292	296	291	285	282
	Edh at Noon		24	30	35	37	33	33	33	33	28	24	21	21
(44) All-Sky Solar Radiation	RadAvg		583	869	1268	1646	1852	2198	2302	1945	1520	961	593	441
	RadStd		59	104	112	137	195	196	121	115	136	87	65	51

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (12 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page